

MT-104

Linear Algebra

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Muhammad Ali

m.ali@nu.edu.pk

National University of Computer and Emerging Sciences

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Week 3

Definition of a Vector Space

A **vector space** (or **linear space**) over a field F is a set V along with two operations:

- ▶ **Vector Addition:** A binary operation $+: V \times V \rightarrow V$.
- ▶ **Scalar Multiplication:** A binary operation $\cdot: F \times V \rightarrow V$.

These operations must satisfy the following axioms:

- ▶ **Closure under Addition and Scalar Multiplication:**

For all $u, v \in V$, $u + v \in V$ (closure under addition).

For all $\alpha \in F$ and $v \in V$, $\alpha \cdot v \in V$.

(closure under scalar multiplication)

- ▶ **Associativity of Addition:**

$$(u + v) + w = u + (v + w) \text{ for all } u, v, w \in V.$$

- ▶ **Commutativity of Addition:**

$$u + v = v + u \text{ for all } u, v \in V.$$

- ▶ **Existence of Additive Identity:**

$$\exists 0 \in V \text{ such that } u + 0 = u \text{ for all } u \in V.$$

► **Existence of Additive Inverses:**

For each $u \in V$, there exists $-u \in V$ such that $u + (-u) = 0$.

► **Distributivity of Scalar Multiplication over Vector Addition:**

$$\alpha \cdot (u + v) = \alpha \cdot u + \alpha \cdot v \text{ for all } \alpha \in F \text{ and } u, v \in V.$$

► **Distributivity of Scalar Multiplication over Field Addition:**

$$(\alpha + \beta) \cdot v = \alpha \cdot v + \beta \cdot v \text{ for all } \alpha, \beta \in F \text{ and } v \in V.$$

► **Compatibility of Scalar Multiplication with Field Multiplication:**

$$(\alpha\beta) \cdot v = \alpha \cdot (\beta \cdot v) \text{ for all } \alpha, \beta \in F \text{ and } v \in V.$$

► **Identity Element of Scalar Multiplication:**

$$1 \cdot v = v \text{ for all } v \in V \text{ (where } 1 \text{ is the multiplicative identity in } F\text{)}.$$

Examples of Vector Spaces

- ▶ **Euclidean Space:** $\mathbb{R}^n$ with standard operations.
- ▶ **Polynomial Space:** The set of all polynomials with real coefficients.
- ▶ **Function Space:** The set of all continuous functions from $\mathbb{R}$ to $\mathbb{R}$.

$\mathbb{R}^2$ as a Vector Space

To show that $\mathbb{R}^2$ is a vector space, we need to verify the following axioms using the standard operations of vector addition and scalar multiplication.

Vector Space Axioms:

- **Closure under Addition:** For all $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$,

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \in \mathbb{R}^2$$

- **Closure under Scalar Multiplication:** For all $(x, y) \in \mathbb{R}^2$ and $\alpha \in \mathbb{R}$,

$$\alpha \cdot (x, y) = (\alpha x, \alpha y) \in \mathbb{R}^2$$

- **Associativity of Addition:**

$$((x_1, y_1) + (x_2, y_2)) + (x_3, y_3) = (x_1, y_1) + ((x_2, y_2) + (x_3, y_3))$$

► **Commutativity of Addition:**

$$(x_1, y_1) + (x_2, y_2) = (x_2, y_2) + (x_1, y_1)$$

► **Existence of Additive Identity:**

$$\exists (0, 0) \text{ such that } (x, y) + (0, 0) = (x, y)$$

► **Existence of Additive Inverses:**

For each $(x, y) \in \mathbb{R}^2$,

there exists $(-x, -y)$ such that $(x, y) + (-x, -y) = (0, 0)$

► **Distributivity of Scalar Multiplication over Vector Addition:**

$$\alpha \cdot [(x_1, y_1) + (x_2, y_2)] = \alpha \cdot (x_1, y_1) + \alpha \cdot (x_2, y_2)$$

► **Distributivity of Scalar Multiplication over Field Addition:**

$$(\alpha + \beta) \cdot (x, y) = \alpha \cdot (x, y) + \beta \cdot (x, y)$$

► **Compatibility of Scalar Multiplication with Field Multiplication:**

$$(\alpha\beta) \cdot (x, y) = \alpha \cdot (\beta \cdot (x, y))$$

► **Identity Element of Scalar Multiplication:**

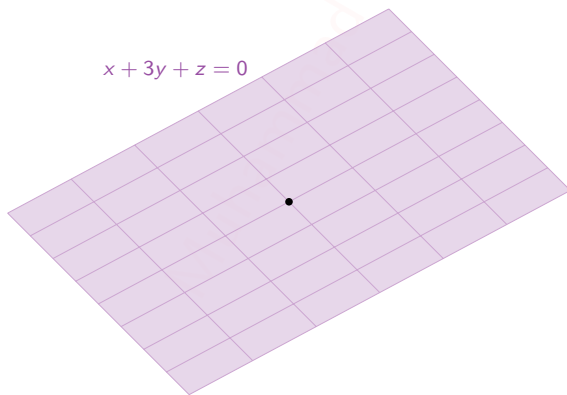
$$1 \cdot (x, y) = (x, y) \text{ where } 1 \text{ is the multiplicative identity in } \mathbb{R}$$

Motivation

Now, we will discuss **subspaces** of $\mathbf{R}^n$.

A subspace turns out to be the same as a span, except we don't know *which* vectors it's the span of.

This arises naturally when you have, say, a plane through the origin in $\mathbf{R}^3$ which is *not* defined (a priori) as a span, but you still want to say something about it.



Definition of Subspace

Definition

A **subspace** of $\mathbf{R}^n$ is a subset V of $\mathbf{R}^n$ satisfying:

1. The zero vector is in V . "not empty"
2. If u and v are in V , then $u + v$ is also in V . "closed under addition"
3. If u is in V and c is in $\mathbf{R}$, then cu is in V . "closed under $\times$ scalars"

What does this mean?

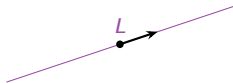
- ▶ If v is in V , then all scalar multiples of v are in V by (3). That is, the line through v is in V .
- ▶ If u, v are in V , then xu and yv are in V for scalars x, y by (3). So $xu + yv$ is in V by (2). So $\text{Span}\{u, v\}$ is contained in V .
- ▶ Likewise, if $v_1, v_2, \dots, v_n$ are all in V , then $\text{Span}\{v_1, v_2, \dots, v_n\}$ is contained in V .

A subspace V contains the span of any set of vectors in V .

Examples

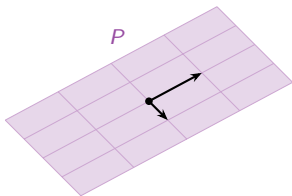
Example

A line L through the origin: this contains the span of any vector in L .



Example

A plane P through the origin: this contains the span of any vectors in P .



Example

All of $\mathbf{R}^n$: this contains 0, and is closed under addition and scalar multiplication.

Example

The subset $\{0\}$: this subspace contains only one vector.

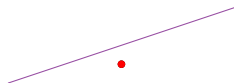
Note these are all pictures of spans! (Line, plane, space, etc.)

Non-Examples

Non-Example

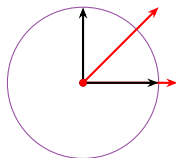
A line L (or any other set) that doesn't contain the origin is not a subspace.

Fails: 1.



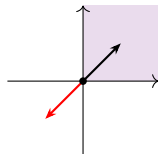
Non-Example

A circle C is not a subspace. Fails: 1,2,3. Think: a circle isn't a "linear space."



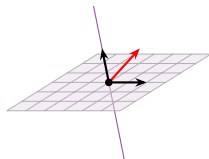
Non-Example

The first quadrant in $\mathbf{R}^2$ is not a subspace. Fails: 3 only.



Non-Example

A line union a plane in $\mathbf{R}^3$ is not a subspace. Fails: 2 only.



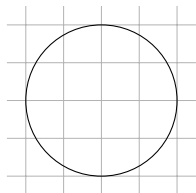
Subsets and Subspaces

They aren't the same thing

A **subset** of $\mathbf{R}^n$ is any collection of vectors whatsoever.

All of the non-examples are still subsets.

A **subspace** is a special kind of subset, which satisfies the three defining properties.



Subset: *yes*

Subspace: *no*

Spans are Subspaces

Theorem

Any $\text{Span}\{v_1, v_2, \dots, v_n\}$ is a subspace.

!!!

Every subspace is a span, and every span is a subspace.

Definition

If $V = \text{Span}\{v_1, v_2, \dots, v_n\}$, we say that V is the subspace **generated by** or **spanned by** the vectors $v_1, v_2, \dots, v_n$.

Check:

1. $0 = 0v_1 + 0v_2 + \dots + 0v_n$ is in the span.
2. If, say, $u = 3v_1 + 4v_2$ and $v = -v_1 - 2v_2$, then

$$u + v = 3v_1 + 4v_2 - v_1 - 2v_2 = 2v_1 + 2v_2$$

is also in the span.

3. Similarly, if u is in the span, then so is cu for any scalar c .

Poll

Is the empty set $\{\}$ a subspace? If not, which property(ies) does it fail?

The zero vector is not contained in the empty set, so it is *not* a subspace.

Question: What is the difference between $\{\}$ and $\{0\}$?

Subspaces

Verification

Let $V = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} \text{ in } \mathbf{R}^2 \mid ab = 0 \right\}$. Let's check if V is a subspace or not.

1. Does V contain the zero vector? $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies ab = 0$ ✓

3. Is V closed under scalar multiplication?

▶ Let $\begin{pmatrix} a \\ b \end{pmatrix}$ be in V .

▶ *This means:* a and b are numbers such that $ab = 0$.

▶ Let c be a scalar. Is $c\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ca \\ cb \end{pmatrix}$ in V ?

▶ *This means:* $(ca)(cb) = 0$.

▶ Well, $(ca)(cb) = c^2(ab) = c^2(0) = 0$ ✓

2. Is V closed under addition?

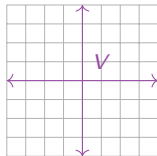
▶ Let $\begin{pmatrix} a \\ b \end{pmatrix}$ and $\begin{pmatrix} a' \\ b' \end{pmatrix}$ be in V .

▶ *This means:* $ab = 0$ and $a'b' = 0$.

▶ Is $\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} a' \\ b' \end{pmatrix} = \begin{pmatrix} a+a' \\ b+b' \end{pmatrix}$ in V ?

▶ *This means:* $(a+a')(b+b') = 0$.

▶ This is not true for all such a, a', b, b' : for instance, $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are in V , but their sum $\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is not in V , because $1 \cdot 1 \neq 0$. ✗



We conclude that V is *not* a subspace. A picture is above. (It doesn't look like a span.)

Column Space and Null Space

An $m \times n$ matrix A naturally gives rise to two subspaces.

Definition

- ▶ The **column space and row space** of A is the subspace of $\mathbf{R}^m$ and $\mathbf{R}^n$ spanned by the columns and rows of A . It is written $\text{Col } A$, $\text{Row}(A)$.
- ▶ The **null space** of A is the set of all solutions of the homogeneous equation $Ax = 0$:

$$\text{Nul } A = \{x \text{ in } \mathbf{R}^n \mid Ax = 0\}.$$

This is a subspace of $\mathbf{R}^n$.

The column and row space is defined as a span, so we know it is a subspace.

Check that the null space is a subspace:

1. 0 is in $\text{Nul } A$ because $A0 = 0$.
2. If u and v are in $\text{Nul } A$, then $Au = 0$ and $Av = 0$. Hence
$$A(u + v) = Au + Av = 0,$$
so $u + v$ is in $\text{Nul } A$.
3. If u is in $\text{Nul } A$, then $Au = 0$. For any scalar c , $A(cu) = cAu = 0$. So cu is in $\text{Nul } A$.

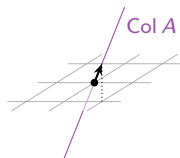
Column Space and Null Space

Example

$$\text{Let } A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Let's compute the column space:

$$\text{Col } A = \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}.$$



This is a line in $\mathbf{R}^3$.

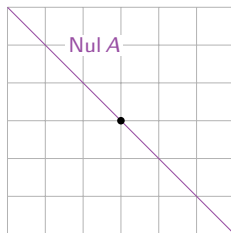
Let's compute the null space:

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ x+y \\ x+y \end{pmatrix}.$$

This is zero if and only if $x = -y$. So

$$\text{Nul } A = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \text{ in } \mathbf{R}^2 \mid y = -x \right\}.$$

This defines a line in $\mathbf{R}^2$:



The Null Space is a Span

The column space of a matrix A is defined to be a span (of the columns).

The null space is defined to be the solution set to $Ax = 0$. It is a subspace, so it is a span.

Question

How to find vectors which span the null space?

Answer: Parametric vector form! We know that the solution set to $Ax = 0$ has a parametric form that looks like

$$x_3 \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -2 \\ 3 \\ 0 \\ 1 \end{pmatrix} \quad \begin{array}{l} \text{if, say, } x_3 \text{ and } x_4 \\ \text{are the free} \\ \text{variables. So} \end{array} \quad \text{Nul } A = \text{Span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 3 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Refer back to the slides for §1.5 (Solution Sets).

Note: It is much easier to define the null space first as a subspace, then find spanning vectors *later*, if we need them. This is one reason subspaces are so useful.

The Null Space is a Span

Example, revisited

Find vector(s) that span the null space of $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$.

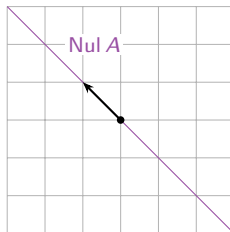
The reduced row echelon form is $\begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$.

This gives the equation $x + y = 0$, or

$$\begin{array}{l} x = -y \\ y = y \end{array} \xrightarrow{\text{parametric vector form}} \begin{pmatrix} x \\ y \end{pmatrix} = y \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

The null space is

$$\text{Nul } A = \text{Span} \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}.$$



Subspaces

Summary

How do you check if a subset is a subspace?

- ▶ Is it a span? Can it be written as a span?
- ▶ Can it be written as the column space of a matrix?
- ▶ Can it be written as the null space of a matrix?
- ▶ Is it all of $\mathbf{R}^n$ or the zero subspace $\{0\}$?
- ▶ Can it be written as a type of subspace that we'll learn about later (eigenspaces, ...)?

If so, then it's automatically a subspace.

If all else fails:

- ▶ Can you verify directly that it satisfies the three defining properties?

Basis of a Subspace

What is the *smallest number* of vectors that are needed to span a subspace?

Definition

Let V be a subspace of $\mathbf{R}^n$. A **basis** of V is a set of vectors $\{v_1, v_2, \dots, v_m\}$ in V such that:

1. $V = \text{Span}\{v_1, v_2, \dots, v_m\}$, and
2. $\{v_1, v_2, \dots, v_m\}$ is linearly independent.

The number of vectors in a basis is the **dimension** of V , and is written $\dim V$.

Note the big
red border here

Why is a basis the smallest number of vectors needed to span?

Recall: *linearly independent* means that every time you add another vector, the span gets bigger.

Hence, if we remove any vector, the span gets *smaller*: so any smaller set can't span V .

Important

A subspace has *many different* bases, but they all have the same number of vectors (see the exercises in §2.9).

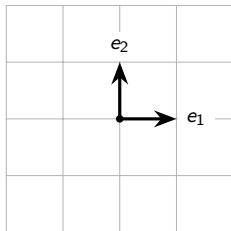
Bases of $\mathbf{R}^2$

Question

What is a basis for $\mathbf{R}^2$?

We need two vectors that *span* $\mathbf{R}^2$ and are *linearly independent*. $\{e_1, e_2\}$ is one basis.

1. They span: $\begin{pmatrix} a \\ b \end{pmatrix} = ae_1 + be_2$.
2. They are linearly independent because they are not collinear.

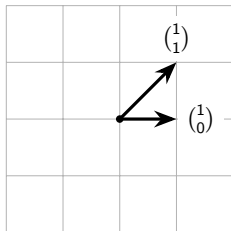


Question

What is another basis for $\mathbf{R}^2$?

Any two nonzero vectors that are not collinear. $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ is also a basis.

1. They span: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has a pivot in every row.
2. They are linearly independent: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has a pivot in every column.



Bases of $\mathbf{R}^n$

The unit coordinate vectors

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \quad \dots, \quad e_{n-1} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \end{pmatrix}, \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

are a basis for $\mathbf{R}^n$.  The identity matrix has columns $e_1, e_2, \dots, e_n$.

1. They span: I_n has a pivot in every row.
2. They are linearly independent: I_n has a pivot in every column.

In general: $\{v_1, v_2, \dots, v_n\}$ is a basis for $\mathbf{R}^n$ if and only if the matrix

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}$$

has a pivot in every row and every column, i.e. if A is *invertible*.

Basis of a Subspace

Example

Example

Let

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ in } \mathbf{R}^3 \mid x + 3y + z = 0 \right\} \quad \mathcal{B} = \left\{ \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} \right\}.$$

Verify that $\mathcal{B}$ is a basis for V .

0. In V : both vectors are in V because

$$-3 + 3(1) + 0 = 0 \quad \text{and} \quad 0 + 3(1) + (-3) = 0.$$

1. Span: If $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ is in V , then $y = -\frac{1}{3}(x + z)$, so

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = -\frac{x}{3} \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix} - \frac{z}{3} \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}.$$

2. Linearly independent:

$$c_1 \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} = 0 \implies \begin{pmatrix} -3c_1 \\ c_1 + c_2 \\ -3c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \implies c_1 = c_2 = 0.$$

Basis for Nul A

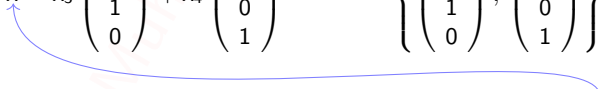
Fact

The vectors in the parametric vector form of the general solution to $Ax = 0$ always form a basis for $\text{Nul } A$.

Example

$$A = \begin{pmatrix} 1 & 2 & 0 & -1 \\ -2 & -3 & 4 & 5 \\ 2 & 4 & 0 & -2 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 0 & -8 & -7 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

parametric vector form $\xrightarrow{\text{~~~~~}}$

$$x = x_3 \begin{pmatrix} 8 \\ -4 \\ 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 7 \\ -3 \\ 0 \\ 1 \end{pmatrix} \xrightarrow{\text{basis of Nul } A} \left\{ \begin{pmatrix} 8 \\ -4 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 7 \\ -3 \\ 0 \\ 1 \end{pmatrix} \right\}$$


1. The vectors span $\text{Nul } A$ by construction (every solution to $Ax = 0$ has this form).
2. Can you see why they are linearly independent? (Look at the last two rows.)

Basis for Col A

Fact

The *pivot columns* of A always form a basis for Col A .

Warning: I mean the pivot columns of the *original* matrix A , not the row-reduced form. (Row reduction changes the column space.)

Example

$$A = \begin{pmatrix} 1 & 2 & 0 & -1 \\ -2 & -3 & 4 & 5 \\ 2 & 4 & 0 & -2 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 0 & -8 & -7 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

pivot columns = basis $\longleftrightarrow$ pivot columns in rref

So a basis for Col A is

$$\left\{ \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} \right\}.$$

Why?